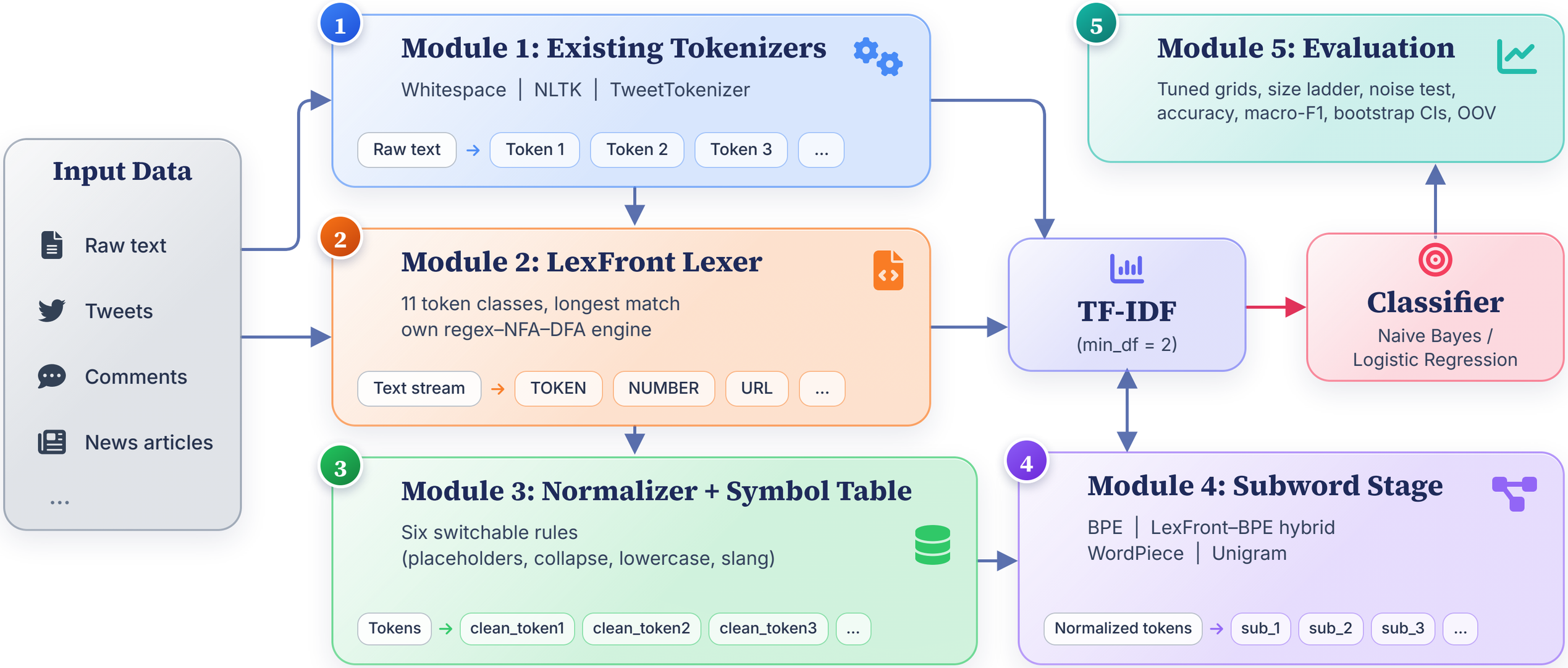

Input Data
Raw text
Tweets
Comments
News articles
...
1
Module 1: Existing Tokenizers
Whitespace | NLTK | TweetTokenizer
Raw text → Token 1 Token 2 Token 3 ...
2
Module 2: LexFront Lexer
11 token classes, longest match
own regex–NFA–DFA engine
Text stream → TOKEN NUMBER URL ...
3
Module 3: Normalizer + Symbol Table
Six switchable rules
(placeholders, collapse, lowercase, slang)
Tokens → clean_token1 clean_token2 clean_token3 ...
TF-IDF
(min_df = 2)
4
Module 4: Subword Stage
BPE | LexFront–BPE hybrid
WordPiece | Unigram
Normalized tokens → sub_1 sub_2 sub_3 ...
5
Module 5: Evaluation
Tuned grids, size ladder, noise test,
accuracy, macro-F1, bootstrap CIs, OOV
Classifier
Naive Bayes /
Logistic Regression